A practical route to fast 3D RDE CFD

Decision brief | 24 October 2026 | 3d-modelling | Proposal only; no new solver work or compute jobs

Recommend PeleC on Modal GPUs, conditionally. Reuse the existing C++/AMReX/PelePhysics integration; retain installed detonationFoam as the CPU fallback. Both have zero license fees. Our H100 execution works, but the current planar control is 15-18% too fast and fails reaction-structure checks. Full GRI chemistry is a reference, not a demonstrated fast production recipe. Current evidence.

Target milliseconds, then slow the movie. Propose 0-2 ms of simulated startup/circulation initially, extending to 5-8 ms if settling requires it. At our reference speed a wave circles in about 0.126 ms: 2 ms covers about 16 revolutions. One physical second costs roughly 500 times a 2 ms run at unchanged workload. A 10-30 second animation can show those milliseconds; it must display the actual physical clock. Two-hour completion is a target, not established feasibility.

First model: gaseous methane/oxygen, thermally perfect ideal-gas species, FP64, full 3D annulus and core-end expansion. Preserve temperature-dependent thermodynamics and finite-rate kinetics; omit droplets, radiation and solid-wall thermal evolution initially. An equivalent premixed inlet and finite ignition pulse can show wave establishment in that idealized model. Predicting the real injection-to-ignition sequence requires separate jets and the predetonator. Mundt is the experimental reference; the CFD papers below are separate studies.

Frameworks assessed for this RDE task

Framework / source	License / compute	Fit and adaptation judgment
PeleC	Free; GPU + CPU	First choice: existing build; AMR and stiff chemistry. Native geometry/propagation still need checks. Published RDE use.
detonationFoam	Free; CPU/MPI	Lowest migration effort; installed detonation-specific solver. No verified GPU path or 3D AMR in our build.
charLES	Paid; GPU	Best directly relevant fast benchmark found; extended flamelet chemistry. Quote required; not a cheap-library assumption.
CONVERGE 6	Paid; CPU + GPU	Strong RDE/CAD workflow. V6 advertises multi-GPU detailed chemistry; matching RDE performance remains unverified.
Fluent	Paid; CPU + GPU	2026 R1 GPU supports stiff chemistry; historical NETL RDE results were CPU. License and matching-case checks needed.
STAR-CCM+	Paid; RDE CPU evidence	Published RDE work; no verified low-cost, fast GPU recipe for our chemistry. More migration than current choices.
UMdetFOAM-GPU	Research GPU code	Actual full-geometry RDE GPU work; public reproducible distribution/access not established. Cannot assume stock OpenFOAM provides it.
AMROC	Open framework; CPU/MPI	Strong 3D RDE/AMR evidence. Higher setup effort; exact research extensions/access need confirmation.
BYCFoam	OpenFOAM research code	RDE-specific, Cantera, AMR/load balancing. Verify source availability; no demonstrated end-to-end GPU path.
CDSFoam / DeepFlame	Public source; mixed CPU/GPU	RDE example and CVODE. GPU low-Mach or neural chemistry features do not certify a GPU detonation solver.
AHFM	Research; CPU evidence	Realistic RDRE benchmark: 1 million core-hours/ms. Access and expense make it unsuitable for this first budget.
MFC	Free; GPU + CPU	Reserve only: reacting/shock examples; no matching full-engine methane RDE demonstration verified. New setup burden.

Not shortlisted: GEMS, JENRE and DET3D/DUSF require research-code access/setup not established here; FLASH and S3D add substantial combustion-case adaptation. Loci/CHEM nozzle studies do not establish a ready reacting engine case. COMSOL has no verified advantage for this shock/chemistry workload. No inexpensive commercial subscription was verified that beats the existing open-source route. CONVERGE academic access may be free if eligible; Horizon Bronze licenses start at $18/node-hour before hardware, so paid access needs a demonstrated time saving.

Budget, assumptions and the next decision

Costs below are allocation arithmetic, not predicted engine runtimes. Overnight means 12 hours here.

Available to us: saved October 22 evidence confirms one H100 80 GB and an eight-core Modal allocation; live authentication/rates work today. Modal documents up to eight GPUs on one host and has published 64-CPU containers, but our eight-GPU/64-core allocation, topology and account ceiling remain unverified. Settings expose no quota; browser requires sign-in. Local Mac: 4 physical cores, 16 GiB RAM, Intel graphics. Azure is enabled, but East US/West US 2 usage queries returned no quota records; Hopper SSH did not resolve. Neither is counted as available capacity.

Candidate allocation	USD/hour	2 hours	12 hours
2 H100 + 8 CPU + 32 GiB host RAM	$8.53	$17.07	$102.41
4 H100 + 16 CPU + 64 GiB host RAM	$17.07	$34.14	$204.83
8 H100 + 32 CPU + 128 GiB host RAM	$34.14	$68.28	$409.65
CPU fallback: 64 physical cores + 128 GiB	$4.05	$8.10	$48.61

Live Modal Function rates: H100 $3.95/GPU-h; A100-80GB $2.50; CPU $0.04730/core-h; RAM $0.008/GiB-h. Excludes build time, storage, transfers, retries and tax; actual CPU/RAM usage can exceed requests. H100 is cheaper per completed job only if it is >1.58x faster than A100 at equal GPU count and GPU-dominated cost. Pricing; GPU limits; CPU resources.

What the papers justify

Argonne / Pal 2021 used CONVERGE 2.4, separate injectors, a predetonator, LES/AMR and 12 species/37 reactions: reuse the modeling strategy, not a presumed runtime. NETL / Strakey 2022 used Fluent and 12 species/38 reactions; coarsening 200 to 400 micrometres changed two waves to one. Senga 2024 / charLES reports 6 ms in 11.2/19.36 hours on four A100s for 3.56/7.58M cells with extended flamelet chemistry; wave speeds 1224/1113 m/s versus 1668 measured. Timing followed a startup/restart procedure, not an end-to-end cold start. This supports an overnight possibility, not guaranteed fidelity or PeleC throughput.

Assumptions worth trading - and what each costs scientifically

Choice	Possible saving	Accuracy / scope sacrificed
Coarsen every spatial direction by 2x	~8x fewer cells; ~16x less explicit flow work at CFL-limited dt.	No universal % error. Peaks, ignition and even wave count can change; test mesh and timestep separately.
AMR around shocks, induction and reaction	If 10% is fine and the rest has 2x spacing: idealized ~4.7x fewer cells than uniformly fine; overhead reduces saving.	Tags must retain cold induction gas and contacts. Our annulus + expansion: ~27.6M fluid-equivalent cells at 0.25 mm; this is not approved resolution.
Published skeletal methane mechanism	53 to 12 species gives ~3.4x smaller basic state; 325 to 38 reactions gives ~8.6x fewer rate terms, not equal total speedup.	Compare ignition, heat release and detonation structure over operating states. Our FFCM1_Red already disagrees with GRI ignition; GRI also has fit-range limitations.
Premixed equivalent inlet; full 360-degree annulus	Omit 144 resolved injector holes/plenums; saving must be measured.	Loses jet mixing and injector recovery dynamics. Can show an idealized engine, not predict injector design or actual cold-start timing.
Reactive Euler, slip/adiabatic walls first	Avoid transport, wall layers and thermal coupling; no honest fixed factor.	Omits viscous/thermal losses and turbulent mixing. Add Navier-Stokes/LES later; low-Mach or steady flow cannot represent rotating detonations.
Finite ignition pulse; 2 ms observation; sparse output	Shorter duration saves roughly linearly. ~200 frames over 2 ms need not be solver timesteps.	Omitting predetonator DDT loses real initiation prediction. Stop ignition before judging circulation; short runs miss slow mode changes and thermal equilibration.

Proposed next work after review: resolve the existing planar propagation defect using supported native numerics; obtain and screen traceable FFCMy-12 from the RDE literature (not FFCM1_Red). Check refill and finite ignition, then benchmark a representative reacting/geometry case on 1, 2, 4 and optionally 8 GPUs. Select the cheapest allocation meeting a measured forecast: <90 minutes integration plus queue/output margin for the two-hour target, or <10 hours plus margin overnight. Full-engine feasibility remains open; do not launch dependent runs while propagation fails.

Deliverable: case inputs/geometry → native CFD → pressure, heat-release and fresh-gas fields → 3D cutaway animation with physical timestamps and front tracks. Start at a declared injection/ignition state; later resolve the real predetonator if cold-start formation is required. Reuse established chemistry integration: the expensive part includes stiff per-cell ODEs and many tiny shock-resolving steps. Adding another general PDE library will not remove those costs.